

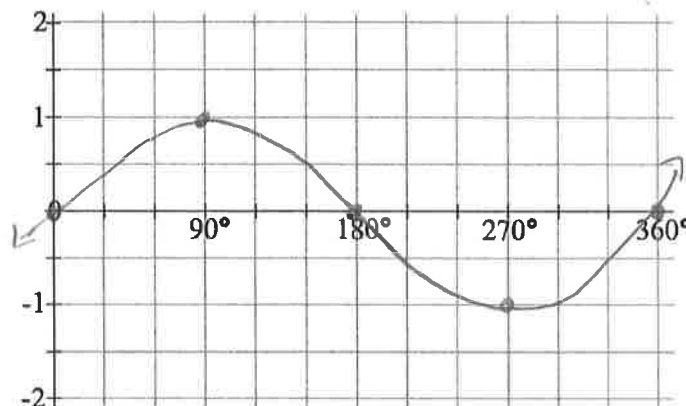
Graphing Sine and Cosine Functions Worksheet

MCR3U

SOLUTIONS

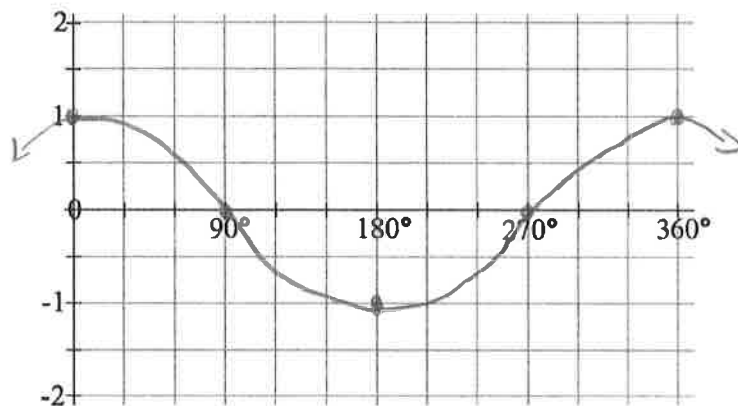
1) Graph the function $y = \sin x$ using key points between 0° and 360° .

x	y
0	0
90	1
180	0
270	-1
360	0



2) Graph the function $y = \cos x$ using key points between 0° and 360° .

x	y
0	1
90	0
180	-1
270	0
360	1



3) Determine the phase shift and the vertical shift of $y = \sin x$.

a) $y = \sin(x - 50^\circ) + 3$

right 50°

up 3

b) $y = 2 \sin(x + 45^\circ) - 1$

left 45°

down 1

4) Determine the phase shift and the vertical shift of $y = \cos x$.

a) $y = -9 \cos(x + 120^\circ) - 5$

left 120°

down 5

b) $y = 12 \cos[5(x - 150^\circ)] + 7$

right 150°

up 7

5) Determine the amplitude, the period, phase shift, vertical shift, maximum and minimum for each of the following.

a) $y = 5 \sin[4(x + 60^\circ)] - 2$

amplitude = 5

$$\text{period} = \frac{360}{4} = 90^\circ$$

shift left 60°

shift down 2

$$\text{max} = 5 - 2 = 3$$

$$\text{min} = -2 - 5 = -7$$

b) $y = 2 \cos[2(x + 150^\circ)] - 5$

amplitude = 2

$$\text{period} = \frac{360}{2} = 180^\circ$$

shift left 150°

shift down 5

$$\text{max} = 2 - 5 = -3$$

$$\text{min} = -5 - 2 = -7$$

c) $y = \frac{1}{2} \sin[\frac{1}{2}(x - 60^\circ)] + 1$

amplitude = $\frac{1}{2}$

$$\text{period} = \frac{360}{0.5} = 720^\circ$$

shift right 60°

shift up 1

$$\text{max} = \frac{1}{2} + 1 = \frac{3}{2} \text{ or } 1.5$$

$$\text{min} = 1 - \frac{1}{2} = \frac{1}{2} \text{ or } 0.5$$

d) $y = 0.8 \cos[3.6(x - 40^\circ)] - 0.4$

amplitude = 0.8

$$\text{period} = \frac{360}{3.6} = 100^\circ$$

shift right 40°

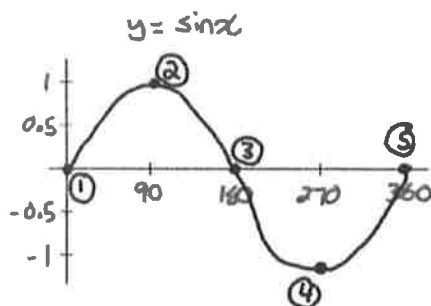
shift down 0.4

$$\text{max} = 0.8 - 0.4 = 0.4$$

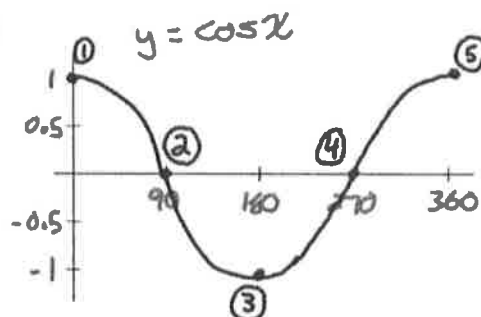
$$\text{min} = -0.4 - 0.8 = -1.2$$

Answers

1)



2)



3) a) phase shift: right 50°
vertical shift: up 3 units

b) phase shift: left 45°
vertical shift: down one unit

4) a) phase shift: left 120°
vertical shift: down 5 units

b) phase shift: right 150°
vertical shift: up 7 units

5) a) amplitude: 5 period: 90° phase shift: left 60°
vertical shift: down 2 units max: 3 min: -7

b) amplitude: 2 period: 180° phase shift: left 150°
vertical shift: down 5 units max: -3 min: -7

c) amplitude: $\frac{1}{2}$ period: 720° phase shift: right 60°
vertical shift: up 1 unit max: 1.5 min: 0.5

d) amplitude: 0.8 period: 100° phase shift: right 40°
vertical shift: down 0.4 units max: 0.4 min: -1.2